

JEE MAIN REVISION MASTER: PRACTICAL ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Practical Organic Chemistry — Applied Cases, Reactions & Traps	3 PYQs	Applied Topics
Standard	Practical Organic Chemistry — Core Principles & Governing Laws	0 PYQs	Core Principle
Standard	Practical Organic Chemistry — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Practical Organic Chemistry — Applied Cases, Reactions & Traps	$\text{C}_6\text{H}_5\text{NH}_2$, H_2SO_4 , $\text{CH}_3\text{CH}_2\text{CHO}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Practical Organic Chemistry — Applied Cases, Reactions & Traps (Moderate Yield — 3 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Statement-I: Nitrogen in pyridine cannot be estimated by Kjeldahl's method. Statement-II: Nitrogen in pyridine changes to ammonium sulphate ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Match List-I with List-II and select the correct options List-I (Functional group) List-II (Functional group reagent) (a) (i) Neutral ferric..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "In the preparation of sodium fusion extract, the purpose of fusing organic compound with a piece of sodium metal is to..." [Ans: N/A]

3.2 Practical Organic Chemistry — Core Principles & Governing Laws (Standard Yield — 0 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Practical Organic Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Practical Organic Chemistry PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Statement-I: Nitrogen in pyridine cannot be estimated by Kjeldahl's method. Statement-II: Nitrogen in pyridine changes to ammonium sulphate ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Match List-I with List-II and select the correct options List-I (Functional group) List-II (Functional group reagent) (a) (i) Neutral ferric..." [Ans: N/A]

3.3 Practical Organic Chemistry — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Statement-I: Nitrogen in pyridine cannot be estimated by Kjeldahl's method. Statement-II: Nitrogen in pyridine changes to ammonium sulphate ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Match List-I with List-II and select the correct options List-I (Functional group) List-II (Functional group reagent) (a) (i) Neutral ferric..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Statement-I: Nitrogen in pyridine cannot be estimated by Kjeldahl's method. Statement-II: Nitrogen in pyridine changes to ammonium sulphate when heated with conc. H_2SO_4 in Kjeldahl's method. Read the above given statements and choose the correct answer from the given options.

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are false
- (C) Both Statement I and Statement II are true
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: Statement-I is true. In Kjeldahl's method, the nitrogen present in an organic compound is converted into ammonium sulphate, $(\text{NH}_4)_2\text{SO}_4$, by heating with concentrated H_2SO_4 . But this method does not work for all nitrogen-containing compounds. It fails for compounds in which nitrogen is present in certain special forms, such as: nitro compounds azo compounds diazo compounds heterocyclic compounds like pyridine. In pyridine, nitrogen is part of the aromatic ring. This nitrogen does not get converted into ammonium sulphate on heating with concentrated